



United International University
Course: CSE 4783 - Cryptography, Spring 2026
Class Test 3, Set A
Total Marks: 20

Name		ID		Section	A
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1. Discuss the suitability of different cover media (audio, video, text, image) for Steganography. [5]

2. Explain the procedure of *embedding on error histogram*. [5]

3. A receiver obtains the following stego prediction errors from a reversible PEH-based steganography system: $[-3, -1, -2, 0, 1, 1, 2]$. The embedding peaks used were: $P^- = -1$ and $P^+ = 0$.

The extraction rules are:

$$s = 0 \quad \text{if } \tilde{e}_{i,j} = P^+ \text{ or } P^-$$

$$s = 1 \quad \text{if } \tilde{e}_{i,j} = P^+ + 1 \text{ or } P^- - 1$$

- a) Extract the embedded secret bits. [5]
- b) Recover the original prediction errors. [5]